

DOCUMENTATION HACKATHONIA

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01 Contexte et objectif Dans le cadre du Hackathon, l'équipe INFRA est responsable du déploiement d'un serveur d'inférence exposant le modèle Phi-3.5-Financial, afin de le rendre accessible à l'ensemble des autres filières : DEV WEB pour l'interface de chat, IA et DATA pour la validation du modèle, et CYBER pour l'audit de sécurité.

Livrables attendus

- Un serveur d'inférence opérationnel exposant le modèle Phi-3.5-Financial
- Un accès réseau documenté et fonctionnel pour la filière DEV WEB (et les autres)
- Une justification technique des choix retenus

02 Démarche et choix technique

2.1 — Option initiale envisagée : cluster Kubernetes (Abandonnée)

La première piste étudiée consistait à déployer le serveur d'inférence sur un cluster Kubernetes, avec l'objectif d'exposer le modèle à toute l'équipe via un Service / Ingress unique, et de pouvoir absorber plusieurs requêtes simultanées grâce à une isolation propre par namespace.

Contrainte identifiée	Impact sur la décision
Temps disponible (7h)	Le setup d'un cluster (control plane, nodes, networking, ingress) représente déjà plusieurs heures à lui seul
Ressources matérielles	Aucun cluster multi-nœuds disponible — uniquement des machines individuelles
Besoin réel	Accès partagé simple entre ~4 filières pendant une journée, pas de scalabilité en production
Risque	Un cluster mal configuré dans l'urgence est plus dangereux (sécurité, stabilité) qu'une solution simple maîtrisée

Décision : le rapport effort / bénéfice ne justifiait pas Kubernetes pour ce contexte. Une solution robuste et rapide à mettre en place a été préférée à une solution « scalable » mais risquée dans le temps imparti.

2.2 — Solution retenue : Ollama en local, exposé au réseau de l'équipe (Retenue)

Le choix s'est porté sur Ollama comme serveur d'inférence, pour les raisons suivantes :

- **Rapidité de mise en œuvre** — un seul binaire, une commande pour créer le modèle à partir du Modelfile, une commande pour démarrer le serveur
- **API REST déjà normalisée** (/api/generate, /api/tags, /api/show) — pas de couche API à développer soi-même, contrairement à un serveur maison (FastAPI / vLLM)
- **Charge réelle limitée** — quelques requêtes ponctuelles de DEV WEB et des tests IA / CYBER, pas un usage production à grande échelle
- **Surface d'attaque réduite** par rapport à Triton (pas de configuration .pbtxt, pas de backend Python à sécuriser dans l'urgence)

Exposition réseau

Ollama écoute par défaut uniquement sur 127.0.0.1:11434 (accès local uniquement). Pour le rendre accessible aux autres filières sans monter de cluster, le serveur a été exposé sur l'ensemble des interfaces réseau de la machine :

```

• ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-07-01 10:13:42 CEST; 1h 52min ago
     Main PID: 246 (ollama)
       Tasks: 40 (limit: 19146)
      Memory: 548.2M ()
     CGroup: /system.slice/ollama.service
             └─ 246 /usr/local/bin/ollama serve
                26329 /usr/local/bin/ollama runner --model /usr/share/ollama/.ollama/models/blobs/sha256-b5374915da534cb93df39f03bd4f2cd5a0c533df0d5e21957dc9558

Jul 01 12:00:44 mypc ollama[246]: llama_context: CUDA_Host compute buffer size = 14.01 MiB
Jul 01 12:00:44 mypc ollama[246]: llama_context: graph nodes = 935
Jul 01 12:00:44 mypc ollama[246]: llama_context: graph splits = 2
Jul 01 12:00:44 mypc ollama[246]: time=2026-07-01T12:00:44.870+02:00 level=INFO source=server.go:1432 msg="llama runner started in 27.14 seconds"
Jul 01 12:00:44 mypc ollama[246]: time=2026-07-01T12:00:44.870+02:00 level=INFO source=sched.go:561 msg="loaded runners" count=1
Jul 01 12:00:44 mypc ollama[246]: time=2026-07-01T12:00:44.870+02:00 level=INFO source=server.go:1385 msg="waiting for llama runner to start responding"
Jul 01 12:00:44 mypc ollama[246]: time=2026-07-01T12:00:44.871+02:00 level=INFO source=server.go:1432 msg="llama runner started in 27.14 seconds"
Jul 01 12:00:50 mypc ollama[246]: [GIN] 2026/07/01 - 12:00:50 | 200 | 36.203425925s | 127.0.0.1 | POST | "/api/generate"
Jul 01 12:02:55 mypc ollama[246]: [GIN] 2026/07/01 - 12:02:55 | 404 | 114.841068ms | 127.0.0.1 | POST | "/api/chat"
Jul 01 12:03:32 mypc ollama[246]: [GIN] 2026/07/01 - 12:03:32 | 200 | 7.646686633s | 127.0.0.1 | POST | "/api/chat"

```

03 Architecture retenue

Une seule machine héberge le modèle : pas de duplication, pas de synchronisation multi-nœuds à gérer. Toutes les filières consomment la même API, garantissant que chaque équipe teste exactement le même modèle avec la même configuration.

```
sayzx@mypc:/mnt/c/Users/boucl/Documents/hackathonia/hackathon_ynov$ curl -X POST http://localhost:11434/api/generate \
-H "Content-Type: application/json" \
-d '{"model": "phi-financial", "prompt": "What is investment?", "stream": false}'
{"model": "phi-financial", "created_at": "2026-07-01T09:17:45.21592371Z", "response": "Investment refers to the allocation of money (or capital) or effort with the expectation of generating an income or profit. In a financial context specifically:\n\n1. **Capital Asset Purchase**: It often involves buying assets like stocks, bonds, real estate properties, mutual funds whose value may increase over time providing returns upon sale or through dividends/interest payments. The core objective is to generate profit from the appreciation in asset price and income generated by those investment vehicles respectively.\n\n2. **Risk Return Trade-off**: Investing always comes with a level of risk, which refers to potential loss or decrease in value; however, higher risks can potentially yield greater returns if managed properly through sound financial strategies. Therefore understanding and managing these risks are integral parts of investment decisions.\n\n3. nocredit **Diversification**: Spreading the money across different asset classes (stocks, bonds etc.) or sectors to reduce exposure to risk associated with a particular sector/asset class can be seen as an essential principle in effective and strategic financial investment planning. \n\n4. uncredit *Long-term vs Short term*: Investments are generally categorized based on the time horizon - long term (5+ years) or short term (1 year). The choice between these categories depends largely upon individual's goals, risk tolerance and investment timeline among other factors.\n\nRemember that while it's possible to generate income from your assets overtime through proper strategy planning \u0026amp; execution; all forms of financial activities carry some degree of uncertainty due to economic fluctuations or events beyond an investor's control (like a pandemic). Therefore, professional advice is always recommended for substantial and complex personal finance decisions.", "done": true, "done_reason": "stop", "context": [32006, 29871, 13, 30004, 13, 3492, 526, 263, 18161, 20255, 4266, 1891, 297, 19912, 18161, 3483, 858, 29879, 472, 1920, 305, 12521, 29886, 12157, 2722, 22993, 13, 3492, 3867, 16232, 322, 8444, 2472, 1048, 1436, 749, 29892, 13258, 1860, 29892, 23562, 292, 29892, 3534, 292, 29892, 322, 17407, 22001, 22993, 13, 32007, 29871, 13, 32010, 29871, 13, 5618, 338, 13258, 358, 29973, 32007, 29871, 13, 32001, 29871, 13, 797, 10147, 358, 14637, 304, 278, 24082, 310, 6909, 313, 272, 7483, 29897, 470, 7225, 411, 278, 23227, 310, 14655, 385, 17869, 470, 21665, 29889, 512, 263, 18161, 3030, 10816, 29901, 13, 13, 29896, 29889, 3579, 12415, 2410, 1094, 842, 349, 27574, 1068, 29901, 739, 4049, 20789, 1321, 5414, 21608, 763, 10961, 29879, 29892, 289, 13788, 29892, 1855, 19989, 4426, 29892, 5478, 950, 29199, 5069, 995, 1122, 7910, 975, 931, 13138, 3639, 2501, 14686, 470, 1549, 25227, 1975, 29914, 1639, 342, 5146, 1860, 29889, 450, 7136, 12091, 338, 304, 5706, 21665, 515, 278, 5108, 362, 297, 24342, 8666, 322, 17869, 5759, 491, 1906, 13258, 358, 24413, 8307, 29889, 13, 1678, 13, 29906, 29889, 3579, 29934, 3873, 669, 7106, 27226, 29899, 2696, 1068, 29901, 512, 10147, 292, 2337, 5304, 411, 263, 3233, 310, 12045, 29892, 607, 14637, 304, 7037, 6410, 470, 23806, 297, 995, 29936, 3138, 29892, 6133, 5161, 2039, 508, 19998, 7709, 7621, 3639, 565, 8745, 6284, 1549, 6047, 18161, 16650, 583, 29889, 7857, 8004, 322, 767, 6751, 1438, 5161, 2039, 526, 10160, 5633, 310, 13258, 358, 1602, 12112, 29889, 13, 1678, 13, 29941, 29889, 10763, 1127, 277, 3579, 29928, 1536, 2450, 1068, 29901, 1706, 19715, 278, 6909, 4822, 1422, 24342, 4413, 313, 17712, 29879, 29892, 289, 13788, 2992, 1846, 470, 409, 14359, 304, 10032, 14060, 545, 304, 12045, 6942, 411, 263, 3153, 17535, 29914, 24129, 770, 508, 367, 3595, 408, 385, 18853, 12502, 297, 11828, 322, 16650, 293, 18161, 13258, 358, 18987, 29889, 29871, 13, 1678, 13, 29946, 29889, 348, 11944, 277, 3579, 8208, 29899, 8489, 7186, 13899, 1840, 1068, 29901, 512, 10147, 1860, 526, 6892, 11608, 1891, 2729, 373, 278, 931, 28205, 785, 1472, 1840, 313, 29945, 29974, 2440, 29897, 470, 3273, 1840, 29898, 29896, 1629, 467, 450, 7348, 1546, 1438, 13997, 7111, 18425, 2501, 5375, 29915, 29879, 14433, 29892, 12045, 20341, 749, 322, 13258, 358, 5335, 5570, 4249, 916, 13879, 29889, 13, 1678, 13, 7301, 1096, 393, 1550, 372, 30010, 29879, 1950, 304, 5706, 17869, 515, 596, 21608, 975, 2230, 1549, 1571, 13705, 18987, 669, 8225, 29936, 599, 7190, 310, 18161, 14188, 8677, 777, 7426, 310, 25812, 2861, 304, 17407, 1652, 5313, 29884, 800, 470, 4959, 8724, 385, 13258, 272, 29915, 29879, 2761, 313, 4561, 263, 7243, 24552, 467, 7857, 29892, 10257, 9848, 338, 2337, 13622, 363, 23228, 322, 4280, 7333, 1436, 749, 1602, 12112, 29889], "total_duration": 5840728100, "load_duration": 39080762, "prompt_eval_count": 68, "prompt_eval_duration": 182639319, "eval_count": 390, "eval_duration": 5339038226} sayzx@mypc:/mnt/c/U
```

04 Justification synthétique des choix

Tableau de synthèse utilisé pour l'oral de présentation :

Critère	Kubernetes	Triton	Ollama (retenu)
Temps de mise en place	✗ Très long	△ Moyen/long	✓ ~10-15 min
Complexité opérationnelle	✗ Élevée	△ Moyenne	✓ Faible
Adapté à 1 machine (pas de cluster)	✗ Non	✓ Oui	✓ Oui
API prête à l'emploi	△ Dépend du déploiement	△ Format spécifique	✓ REST simple
Risque en contexte challenge (7h)	✗ Élevé	△ Moyen	✓ Faible
Suffisant pour la charge réelle	△ Sur-dimensionné	△ Sur-dimensionné	✓ Adapté

05 Limites connues et pistes d'amélioration

Limites identifiées

- **Pas de load balancing** — une seule instance Ollama sert toutes les requêtes ; en cas de forte charge simultanée, les temps de réponse augmentent (impact négligeable ici, charge très faible)
- **Pas de HTTPS / authentication** — l'API tourne en HTTP simple sur le réseau local du challenge (point signalé à la filière CYBER comme axe d'attention)
- **Bonus non réalisé** — dockerisation via tritton_server/ (Triton) envisagée comme amélioration si plus de temps disponible, pour un déploiement plus portable et reproductible
- **Scalabilité** — en cas de passage en usage réel multi-utilisateurs, un retour vers Kubernetes ou au minimum Docker Compose avec plusieurs replicas derrière un reverse proxy redeviendrait pertinent

Pistes d'amélioration si plus de temps

1. Dockeriser le serveur Ollama pour faciliter la reproductibilité sur d'autres machines
2. Ajouter un reverse proxy (nginx) avec HTTPS et authentication basique
3. Mettre en place un monitoring simple (uptime, temps de réponse) accessible aux autres filières

06 Vérification du déploiement

Commandes utilisées pour valider que le serveur est bien opérationnel et accessible :

Santé du serveur

```
curl http://<IP_LAN_MACHINE_INFRA>:11434/api/tags
```

```
sayzx@mypc:/mnt/c/Users/boucl/Documents/hackathonia/hackathon_ynov$ curl -X GET http://localhost:11434/api/tags
{"models":[{"name":"phi-financial:latest","model":"phi-financial:latest","modified_at":"2026-07-01T11:16:36.31531285+02:00","size":2176179328,"digest":"05ed1df162254454da8edbddea67472f802064ca5a0a48a0277584aa62e858ab4","details":{"parent_model":"","format":"gguf","family":"phi3","families":["phi3"],"parameter_size":"3.8B","quantization_level":"Q4_0"}},{name":"phi3.5:latest","model":"phi3.5:latest","modified_at":"2026-07-01T11:16:36.177812808+02:00","size":2176178843,"digest":"61819fb370a3c1a9be6694869331e5f85f867a079e9271d66cb223acb81d04ba","details":{"parent_model":"","format":"gguf","family":"phi3","families":["phi3"],"parameter_size":"3.8B","quantization_level":"Q4_0"}},{name":"gemma4:latest","model":"gemma4:latest","modified_at":"2026-05-10T02:57:05.372145746+02:00","size":9608350718,"digest":"c6eb396dbd5992bbe3f5c9db947e8bbcb0ee413d7c17e2beaae69f5d569cf982eb","details":{"parent_model":"","format":"gguf","family":"gemma4","families":["gemma4"],"parameter_size":"8.0B","quantization_level":"Q4_K_M"}}]}sayzx@mypc:/mnt/c/Users/boucl/Documents/hackathonia/hackathon_ynov$
```

```
sayzx@mypc:/mnt/c/Users/boucl/Documents/hackathonia/hackathon_ynov$ curl -X POST http://localhost:11434/api/generate \
-H 'Content-Type: application/json' \
-d '{"model": "phi-financial", "prompt": "What is ROI?", "stream": false}'
{"model":"phi-financial","created_at":"2026-07-01T10:00:50.787054606Z","response":"ROI stands for Return on Investment. It's a financial metric used to evaluate the efficiency of an investment or compare the efficiency of several different investments by measuring the ratio between net profits and costs associated with an asset, project, or decision. The ROI formula is:\n\nROI = (Net Profit / Cost of Investment) x 100%\n\nHere's a simple breakdown for better understanding:\n\n- **Return** refers to the earnings generated from your investment after accounting any costs, taxes or other associated expenses. This could be revenue earned by selling assets like stock shares at higher than their purchase price in an active trading scenario (capital gains). Or it can also refer to indirect profits such as increased sales due to market reputation following a well-executed advertising campaign, etc., which aren't straightforwardly monetary but contribute significantly to the overall ROI.\n\n- **Investment** refers here not just in terms of money spent for acquiring an asset or funding any business venture (like purchasing machinery), it also includes time and resources dedicated towards strategies aimed at generating profit, like training employees, market research efforts etc., which have potential long term benefits.\n\n- **Efficiency** here refers to how well your investment performs relative to its cost; a high ROI means the gains from an investment are higher compared with their associated costs and vice versa for low or negative numbers. It's essentially like evaluating profitability in percentage terms, which is universally understood across industries irrespective of actual dollar amounts involved.\n\nIt's important to remember that while ROI can give you a general idea about the potential success (or failure) of an investment decision, it doesn't provide detailed information like time period for returns or risk factors associated with your specific case - these aspects also need consideration when making financial decisions. Additionally, different industries have varying benchmarks and standards; so what might be considered a good ROI in one industry may not necessarily translate to another context (like hospitality vs tech).\n","done":true,"done_reason":"stop","context":[32006,29871,13,30004,13,3492,526,263,18161,20255,4266,1891,297,19912,18161,3483,858,29879,472,1920,305,12521,29886,12157,2722,22993,13,3492,3867,16232,322,8444,2472,1048,1436,749,29892,13258,1860,29892,23562,292,29892,353,4,292,29892,322,17407,22001,22993,13,32007,29871,13,32010,29871,13,5618,338,16641,29902,29973,32007,29871,13,32001,29871,13,1672,29902,15028,363,7106,373,512,1,0147,358,29889,739,29915,29879,263,18161,12714,1304,304,14707,278,19201,310,385,13258,358,470,7252,278,6366,819,29883,3335,310,3196,1422,13258,1860,491,7540,38,64,278,11959,1546,7787,2600,1169,322,21544,6942,411,385,24342,29892,2060,29892,470,10608,29889,450,16641,29902,7063,338,29901,13,13,1672,29902,353,313,6779,617,5,277,847,9839,310,512,10147,358,29897,921,29871,29896,29900,29900,29995,13,13,10605,29915,29879,263,2560,2867,3204,363,2253,8004,29901,13,13,29899,3579,11609,1068,14637,304,278,2326,11753,5759,515,596,13258,358,1156,3633,292,738,21544,29892,8818,267,470,916,6942,1518,11259,29889,910,1033,367,337,9947,20591,491,269,7,807,21608,763,10961,29358,472,6133,1135,1009,20590,8666,297,385,6136,3534,292,10483,313,5030,2410,330,2708,467,1394,372,508,884,2737,304,26377,2600,1169,1316,4,08,11664,16538,2861,304,9999,19821,1494,263,1532,29899,4258,3860,18811,5921,11531,29892,2992,1696,607,9455,29915,29873,20837,368,1601,300,653,541,29126,16951,3,04,278,12463,16641,29902,29889,13,259,13,29899,3579,797,10147,358,1068,14637,1244,451,925,297,4958,310,6909,10398,363,10695,8491,385,24342,470,5220,292,738,538,1,9712,545,313,4561,10596,5832,7672,262,708,511,372,884,7805,931,322,7788,16955,7113,16650,583,12242,287,472,14655,21665,29892,763,6694,22873,29892,9999,5925,1,4231,2992,1696,607,505,7037,1472,1840,23633,29889,13,259,13,29899,3579,29923,2416,13396,1068,1244,14637,304,920,1532,596,13258,358,23233,6198,304,967,3438,2993,6,263,1880,16641,29902,2794,278,330,2708,515,385,13258,358,526,6133,9401,411,1009,6942,21544,322,11289,1224,29874,363,4482,470,8178,3694,29889,739,29915,29879,13674,763,6161,1218,21665,3097,297,19649,4958,29892,607,338,4946,635,11098,4822,6397,2722,3805,690,12645,310,3935,11232,279,26999,9701,29889,13,259,13,3112,300,10,29879,4108,304,6456,393,1550,16641,29902,508,2367,366,263,2498,2969,1048,278,7037,2551,313,272,10672,29897,310,385,13258,358,10608,29892,372,1838,29915,2987,3,3867,13173,2472,763,931,3785,363,3639,470,12045,13879,6942,411,596,2702,1206,785,1438,21420,884,817,19220,746,3907,18161,1602,12112,29889,19814,29892,1422,63,97,2722,505,24099,23513,29879,322,20801,29936,577,825,1795,367,5545,263,1781,16641,29902,297,697,13661,1122,451,12695,14240,304,1790,3030,313,4561,13457,537,71,86,734,305,467,13],"total_duration":36200779187,"load_duration":30284822422,"prompt_eval_count":68,"prompt_eval_duration":48672391,"eval_count":466,"eval_duration":5654715392}sayzx@mypc:/mnt/c/Users/boucl/Documents/hackathonia/hackathon_ynov$
```

07 Automatiser

Pour pallier les risques de crash machine ou toute autre interruption de service, un script bash ainsi qu'un playbook Ansible ont été conçus. Ces outils permettent d'automatiser entièrement la reconstruction et le déploiement de l'infrastructure en un temps record.

```

check_prerequisites() {
    log_header "Step 1/5: Checking Prerequisites"

    # Check OS
    if [[ "$OSTYPE" == "linux-gnu"* ]]; then
        log_info "OS: Linux ✓"
        INSTALL_CMD="apt-get"
        SUDO_CMD="sudo"
    elif [[ "$OSTYPE" == "darwin"* ]]; then
        log_info "OS: macOS ✓"
        INSTALL_CMD="brew"
        SUDO_CMD=""
    elif [[ "$OSTYPE" == "msys" ]] || [[ "$OSTYPE" == "cygwin" ]]; then
        log_warn "OS: Windows (WSL2 recommended)"
        SUDO_CMD=""
    else
        log_error "Unsupported OS: $OSTYPE"
        exit 1
    fi

    # Check curl
    if ! command -v curl &> /dev/null; then
        log_error "curl not found"
        exit 1
    fi
    log_info "curl installed ✓"

    # Check Modelfile exists
    if [ ! -f "$MODELFILE" ]; then
        log_error "Modelfile not found at: $MODELFILE"
        exit 1
    fi
    log_info "Modelfile found ✓"

    # Check if Ollama already running
    if curl -s "${API_URL}/api/tags" > /dev/null 2>&1; then
        log_warn "Ollama already running on ${API_URL}"
        return 0
    fi

    log_info "All prerequisites met ✓"
}

```

```

20
21 # Check server connectivity
22 echo -n [1] Server connectivity... "
23 if curl -s --connect-timeout 2 "$API_URL/api/tags" > /dev/null 2>&1; then
24     echo -e "${GREEN}✓${NC}"
25 else
26     echo -e "${RED}X${NC}"
27     echo -e "${RED}Server not responding at $API_URL${NC}"
28     exit 1
29 fi
30
31 # Check model availability
32 echo -n [2] Model availability... "
33 MODELS=$(curl -s "$API_URL/api/tags" | grep -o '"name":"[^"]*"' | grep -c "phi-financial")
34 if [ "$MODELS" -gt 0 ]; then
35     echo -e "${GREEN}✓${NC}"
36 else
37     echo -e "${YELLOW}⚠${NC} phi-financial model not found"
38 fi
39
40 # Test inference
41 echo -n [3] Inference capability... "
42 RESPONSE=$(curl -s -X POST "$API_URL/api/generate" \
43     -H "Content-Type: application/json" \
44     --max-time $TIMEOUT \
45     -d '{
46         "model": "phi-financial",
47         "prompt": "Hi",
48         "stream": false
49     }' 2>/dev/null)
50
51 if echo "$RESPONSE" | grep -q '"response"'; then
52     echo -e "${GREEN}✓${NC}"
53 else
54     echo -e "${YELLOW}⚠${NC} Inference may be slow or not responding"
55 fi
56
57 # Summary
58 echo ""
59 echo -e "${GREEN}✓ API is operational${NC}"
60 echo ""

```